NYC Citi Bike Operational Intelligence

From Raw Data to AI-Driven Logistics

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COURSE : DATA SCIENCE WITH PYTHON

TEAM:
NITISH KAMLESH CHOUDHARY
MOHAMMED AMAAN SHAIKH
YASHASHWINI SIDRAMAPPA AWATE

The Origin Story

From Local Curiosity to Global Scale

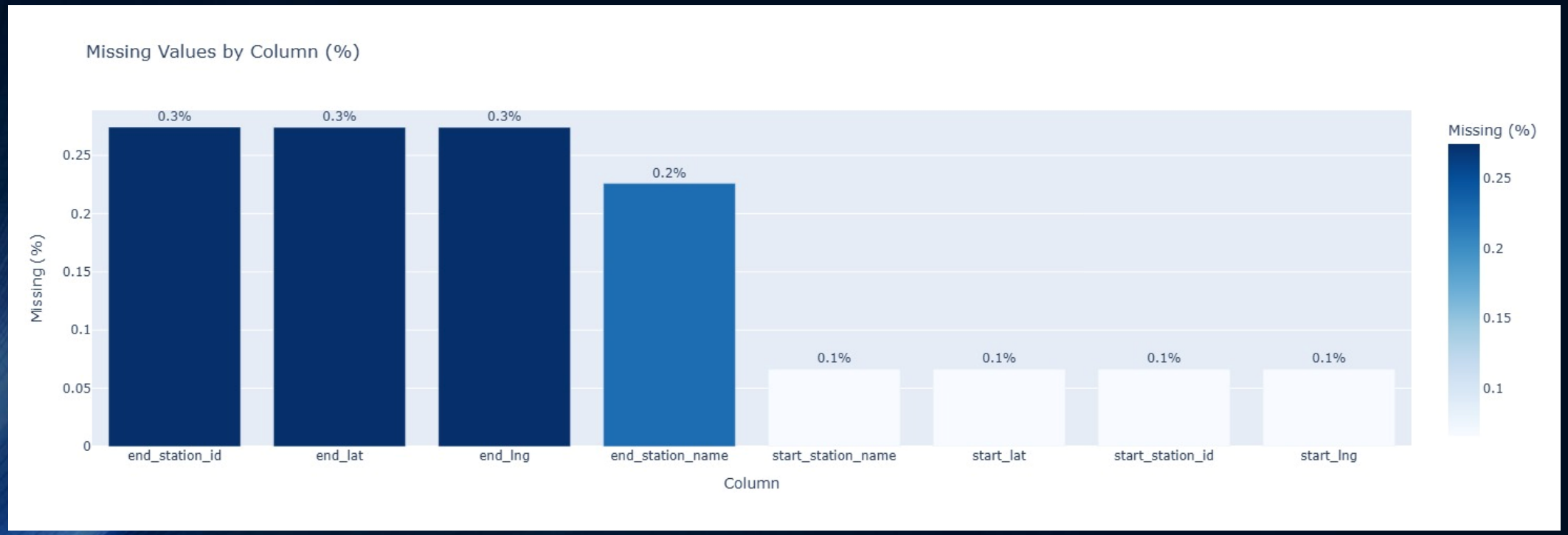
We started with a simple question:

How does a city move? We wanted to model *Magdeburg*, but the data wasn't there.

So we pivoted to NYC, one of the most complex bike network in the world to build a system robust enough to be applied anywhere.

A Glimpse into the Data (Oct 2025)

- **Volume:** 4.7 million trips.
- **Data Cleaning:** Handled missing values by creating a mapping table.
- **Computational Latency:** Basic aggregation queries (e.g., groupby) took exponential time, signaling the need for a streaming architecture rather than in- memory batch processing



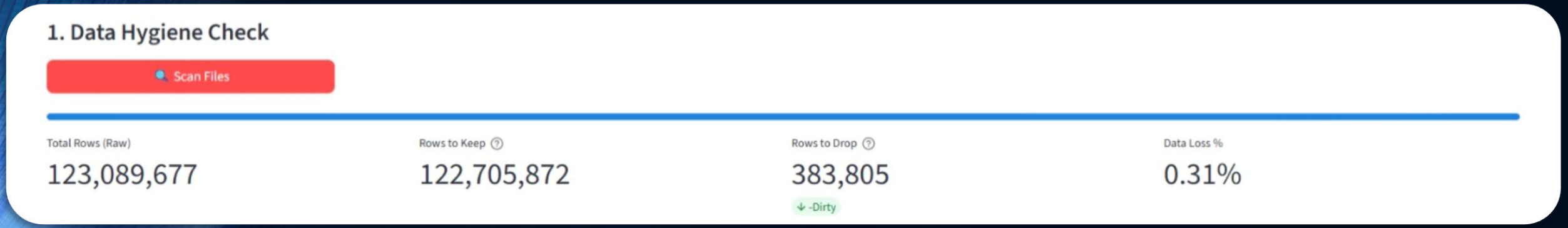
The Data Engineering Architecture

Memory Constraints: Bypassed RAM limits by moving beyond pandas for multi-year, high-frequency log processing.

Strategic Trade-off: Dropped corrupted rows (0.3%) instantly rather than wasting compute on expensive repairs.

Sequential Ingestion: Swapped batch loads for a low-footprint pipeline that processes files one-by- one to free memory.

The Result: Cut processing from hours to minutes by trading <0.4% volume for 100% pipeline stability.



Phase 0 – Data Harmonization

Exploration : We visualized the entire 3-year dataset to find the "Pulse" of the city. The data revealed a distinct Morning/Evening Rush Hour pattern, confirming that Commuters (not Tourists) drive the system's stress.

Narrowing the Scope : Mapping 40M+ rides revealed that Manhattan is the operational engine. To maximize impact, we decided to narrow our Phase 1 diagnostics specifically to this high-volume core.

Fleet Evolution : We uncovered a massive hidden trend: A 70% shift to E-Bikes in 2024. This proved we couldn't just model "Bikes" generically we had to account for electric capacity.

What did you visualize to find the "Pulse" of the city?

We visualized the entire 3-year dataset and found a distinct Morning/Evening Rush Hour pattern, confirming that Commuters drive the system's stress.

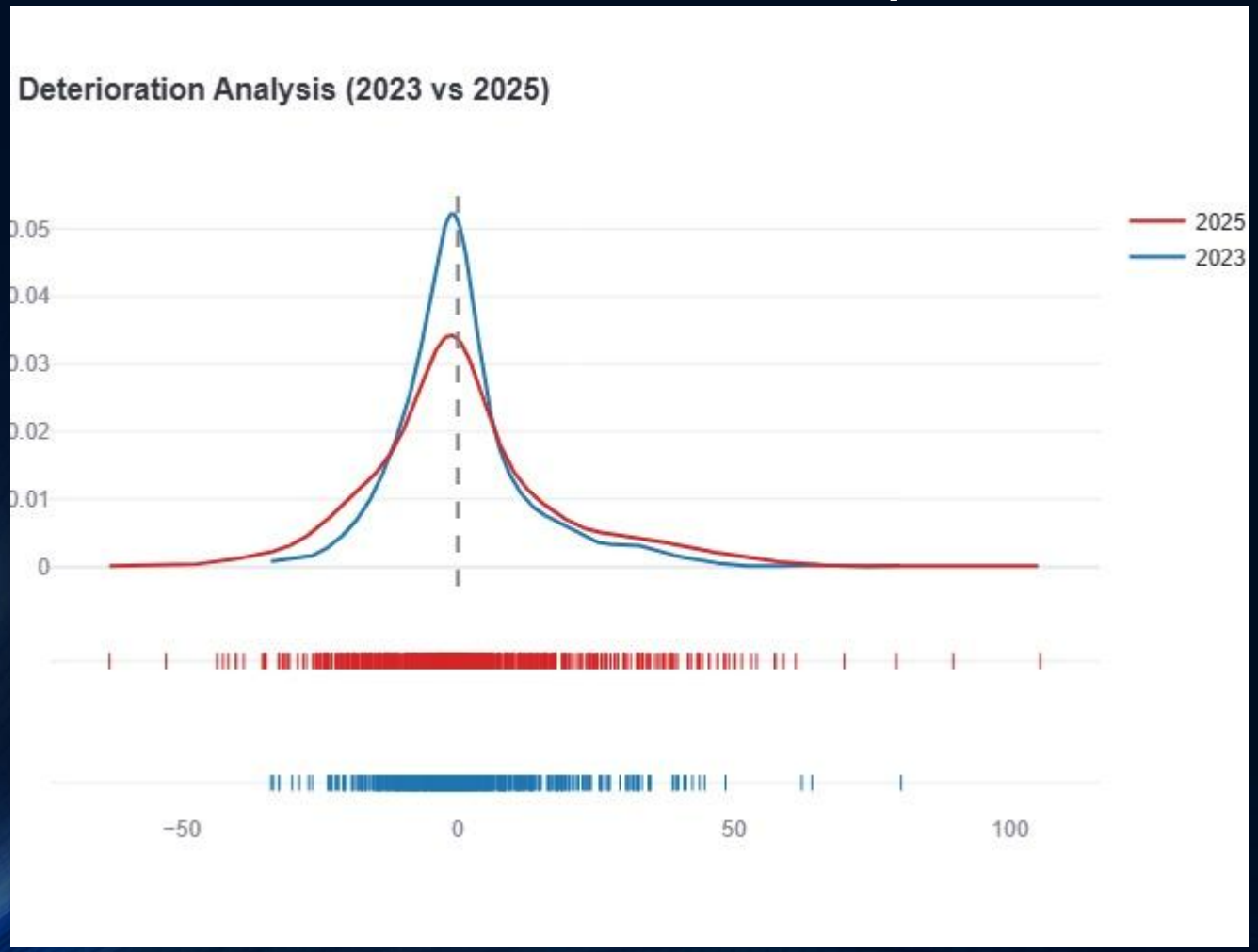
What did mapping 40M+ rides reveal?

Manhattan is the operational engine, so we decided to narrow our Phase 1 diagnostics to this high-volume core.

What massive hidden trend did you uncover?

A 70% shift to E-Bikes in 2024, proving we had to account for electric capacity.

Phase 1 vis+ translons phase 0 to 1



System Deterioration

Why compare years?

2025 data appeared more volatile.

What did the comparison reveal?

The "Flattening Curve" proves that variance is increasing, with more stations hitting critical failure states than ever before.

What does this mean?

Manual logistics can no longer keep up.

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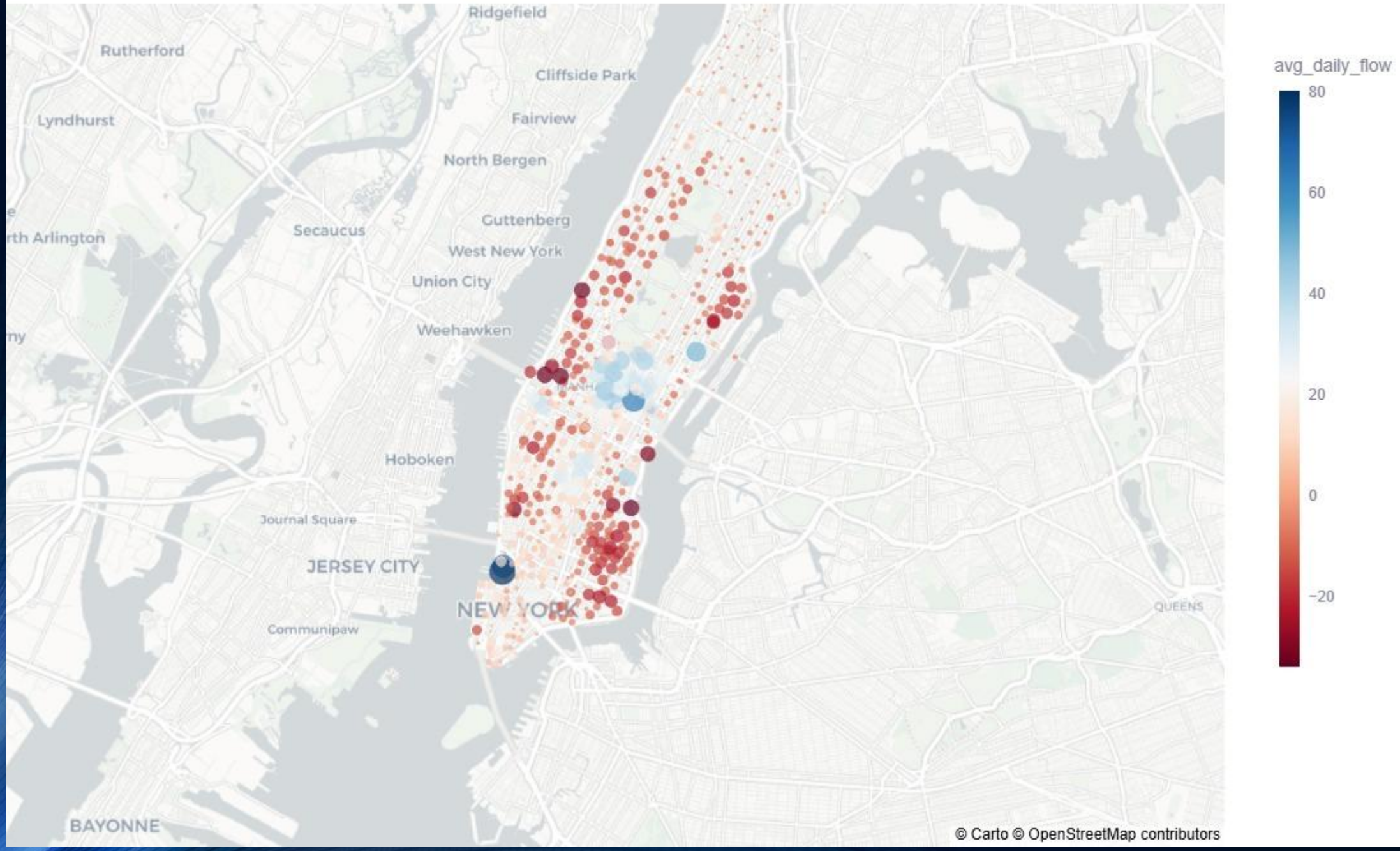
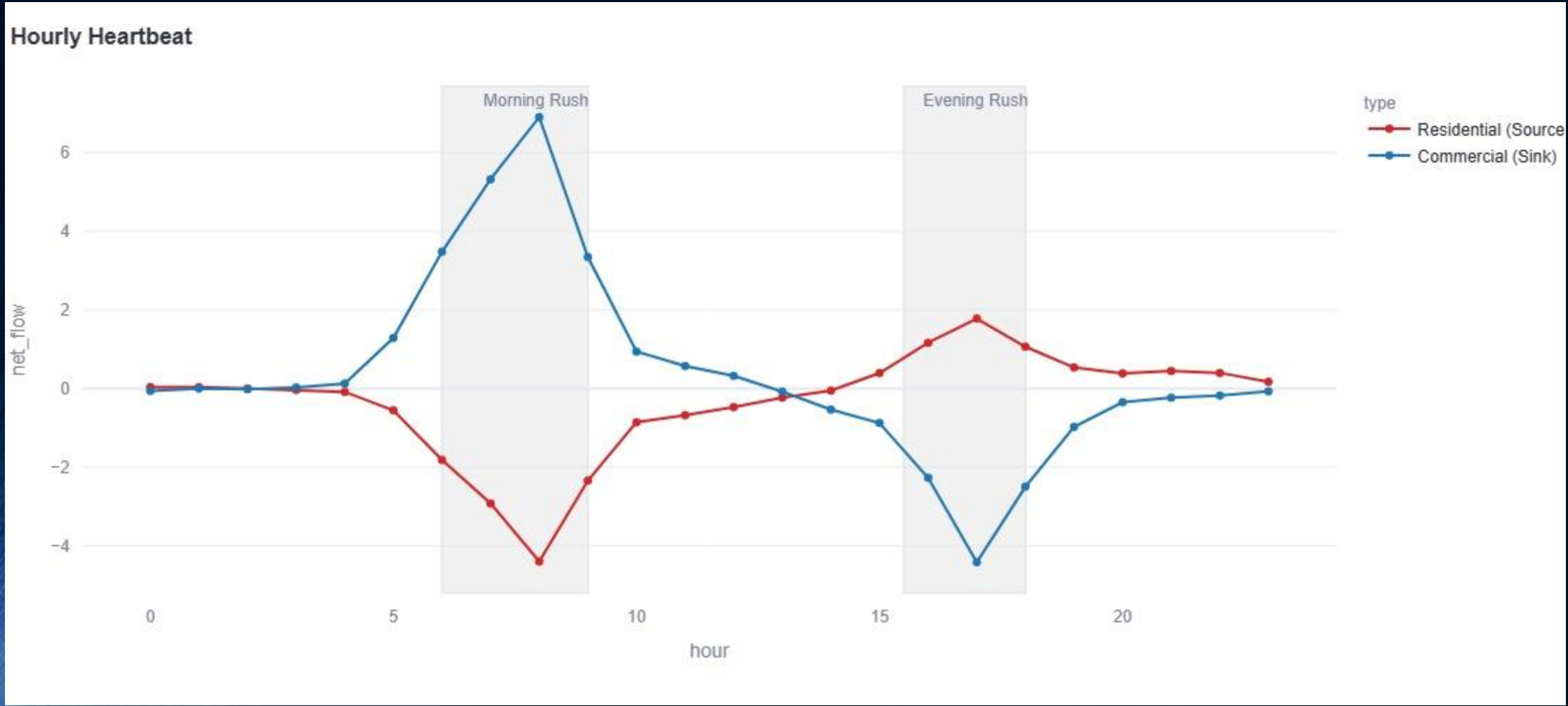
Tidal Wave (2023) - Manhattan Only




Hourly Heartbeat
Morning Rush
Evening Rush
type
Residential (Source
Commercial (Sink)
net_flow
6
4
2
0
−2
−4
0
5
10
15
20
hour

Spatial Profiling of Bike Net Flow

Why Map Net Flow?

High volume doesn't mean high risk. We mapped Net Flow (Returns - Starts) to find Directionality. This identified residential "Sources" (losing bikes) vs. commercial "Sinks" (gaining bikes).

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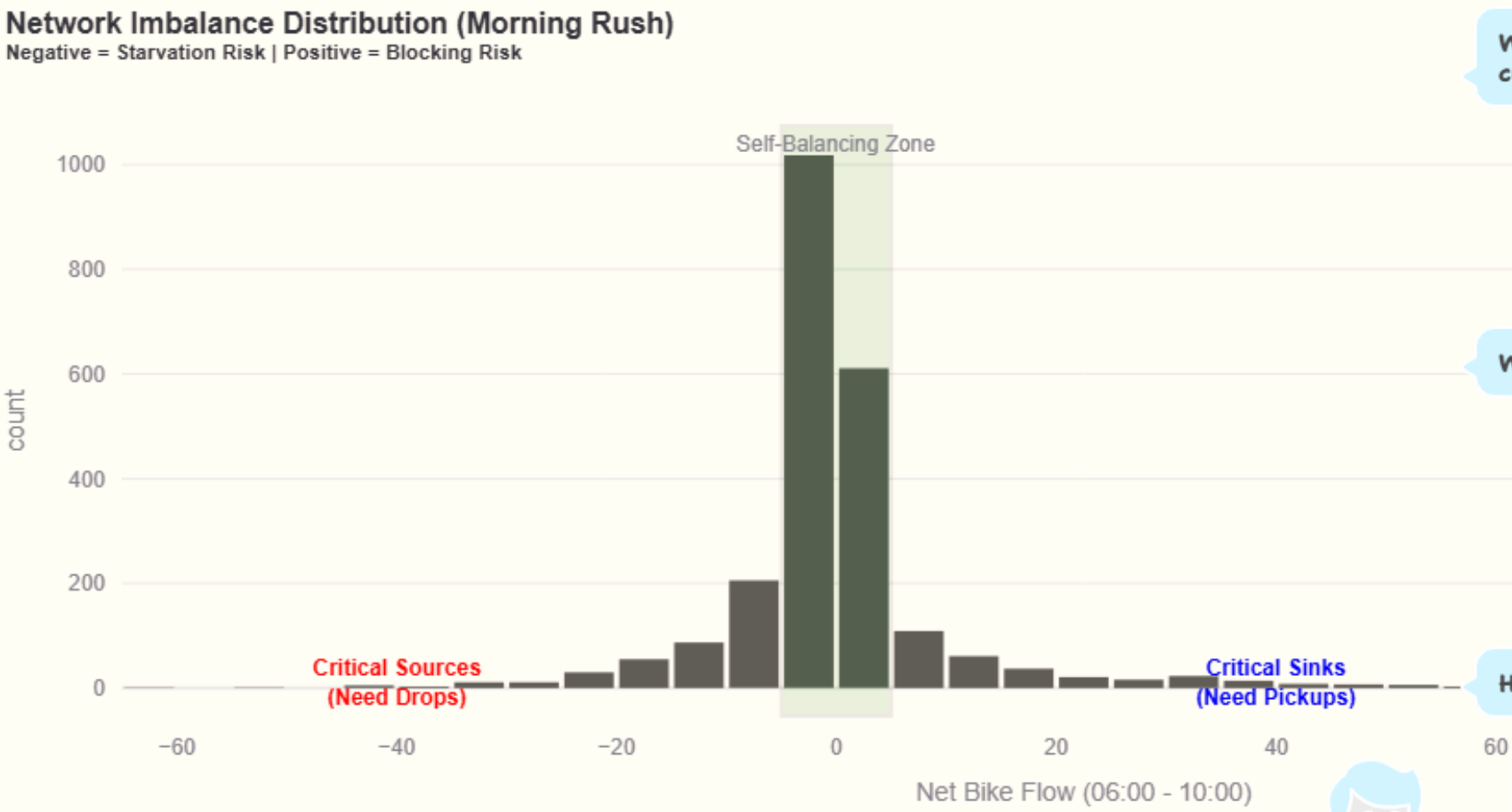
Temporal Profiling of Bike Share System

Why 6-10 AM?

Raw data showed massive volume spikes. We filtered for the Morning Rush to isolate the "Commuter Signal," revealing the exact window where the system is under maximum stress and filtering out tourist noise.

Phase 2



What is the problem with the current bike sharing system?

Critical stations deteriorate faster than trucks can move in morning traffic, and waiting until 8:00 AM to fix the imbalance is too late.

What is the solution?

We shift to a pre-balancing strategy, identifying critical stations and moving inventory overnight to neutralize the imbalance before the first commuter wakes up.

How does the math work?

We target "fat tails" by focusing only on critical "Sources" and "Sinks" and apply a safety buffer to handle unexpected demand spikes.